

1 Support Vector Machines (with Offset)

As we had discussed before, maximum margin linear separators can be found by solving the following primal quadratic programming problem

$$\text{(primal)} \quad \min \frac{1}{2} \|\theta\|^2 \quad \text{subject to } y^{(t)}(\theta \cdot x^{(t)} + \theta_0) \geq 1 \quad (1)$$

If we wish to solve the problem in the dual, i.e., represent the parameters θ in terms of examples as $\theta = \sum_{t=1}^n \alpha_t y^{(t)} x^{(t)}$, we can solve the dual

$$\begin{aligned} \text{(dual)} \quad & \max \sum_{t=1}^n \alpha_t - \frac{1}{2} \sum_{t=1}^n \sum_{t'=1}^n \alpha_t \alpha_{t'} y^{(t)} y^{(t')} (x^{(t)} \cdot x^{(t')}) \\ & \text{subject to } \alpha_t \geq 0, t = 1, \dots, n \quad \text{and} \quad \sum_{t=1}^n \alpha_t y^{(t)} = 0 \end{aligned} \quad (2)$$

where the additional constraint $\sum_{t=1}^n \alpha_t y^{(t)} = 0$ pertains to including the offset parameter θ_0 in the primal. But θ_0 does not appear anywhere in the dual. How do we set it?

After we solve the dual, we can see that $\hat{\alpha}_t > 0$ for some examples (support vectors) and $\hat{\alpha}_t = 0$ for others (non-support vectors). Since for support vectors, the classification constraints must be satisfied with equality, we have that when $\hat{\alpha}_t > 0$,

$$y^{(t)}(\hat{\theta} \cdot x^{(t)} + \hat{\theta}_0) = y^{(t)}\left(\sum_{t'=1}^n \hat{\alpha}_{t'} y^{(t')} (x^{(t')} \cdot x^{(t)}) + \hat{\theta}_0\right) = 1 \quad (3)$$

As $y^{(t)} \in \{-1, 1\}$ or $(y^{(t)})^2 = 1$ we can multiply both sides by $y^{(t)}$ and get

$$\hat{\theta}_0 = y^{(t)} - \left(\sum_{t'=1}^n \hat{\alpha}_{t'} y^{(t')} (x^{(t')} \cdot x^{(t)})\right) \quad \text{subject to } \hat{\alpha}_t > 0 \quad (4)$$

This should hold for all support vectors. However, solving the dual quadratic programming problem numerically does introduce errors and relying on any particular constraint may be unwise. Instead, we can simply take the mean of all $\hat{\theta}_0$ estimates from support vector constraints:

$$\hat{\theta}_0 = \frac{1}{|V|} \sum_{t \in V} \left\{ y^{(t)} - \left(\sum_{t'=1}^n \hat{\alpha}_{t'} y^{(t')} (x^{(t')} \cdot x^{(t)})\right) \right\}, \quad V = \{i | \hat{\alpha}_i > 0\}. \quad (5)$$

2 Support Vector Machines (with Errors)

If the labels for training examples contain errors, finding the maximum margin linear classifier may be problematic. The resulting decision boundary is potentially drastically affected by a single mislabeled point. Consider, for example, how the decision boundary changes by the single positive (perhaps mislabeled) point included in Figure 1b).

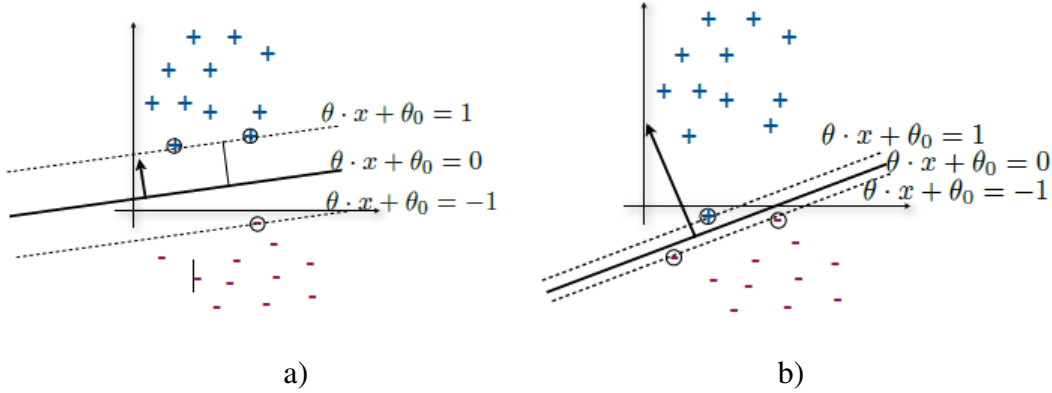


Figure 1: The maximum margin separator is strongly affected by individual points

In order to remedy the situation, we should allow for misclassified points, yet maximize the margin. How do we achieve this? The simplest way is to treat the classification constraints as “soft” constraints rather than hard constraints. In other words, we permit them to be violated but specify the cost of such discrepancies in relation to increasing the margin (decreasing the norm $\|\theta\|$). Specifically, we will add “slack” variables $\xi_t \geq 0$ to the primal optimization problem as follows

$$\text{(primal)} \quad \min \frac{\lambda}{2} \|\theta\|^2 + \sum_{t=1}^n \xi_t \quad (6)$$

$$\text{subject to } y^{(t)}(\theta \cdot x^{(t)} + \theta_0) \geq 1 - \xi_t, \xi_t \geq 0, t = 1, \dots, n \quad (7)$$

We now minimize with respect to the parameters θ , θ_0 as well as the slack variables $\xi_t \geq 0$. Note that we had to introduce a parameter¹ λ (regularization parameter) to balance how much we favour increasing the margin over satisfying the classification constraints. Larger values of λ will push the margin boundaries and potentially the decision boundary past the examples. The margin boundaries are still defined as points satisfying $\theta \cdot x + \theta_0 = 1$ or $\theta \cdot x + \theta_0 = -1$. Figure 2 illustrates the choice of λ and the resulting maximum margin solutions when the examples are still linearly separable.

¹In the literature you will often see a parameter C multiplying the sum of slack variables instead. This is the same formulation so long as $C = 1/\lambda$

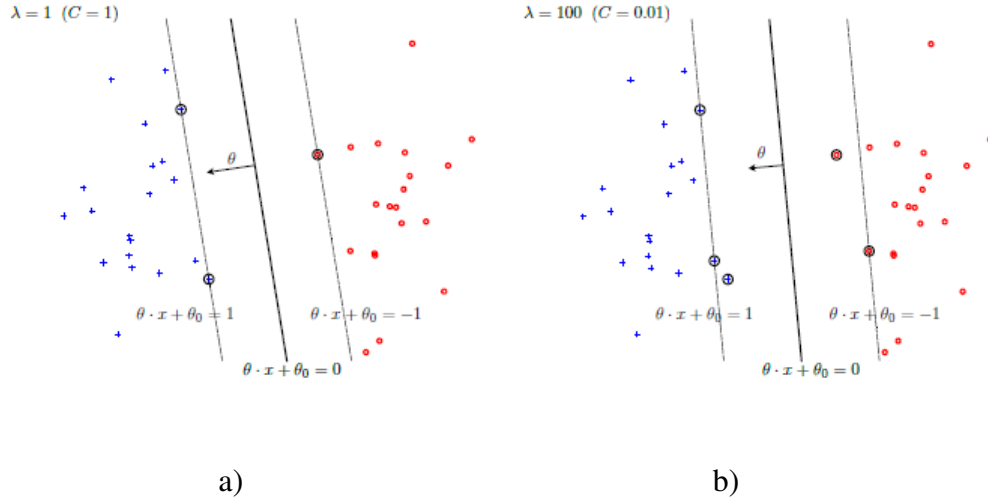


Figure 2: The effect of slack when examples are still linearly separable

The other advantage of the slack variables is that we can now solve problems that are no longer linearly separable. This is illustrated in Figure 3 with different values of the regularization parameter λ .

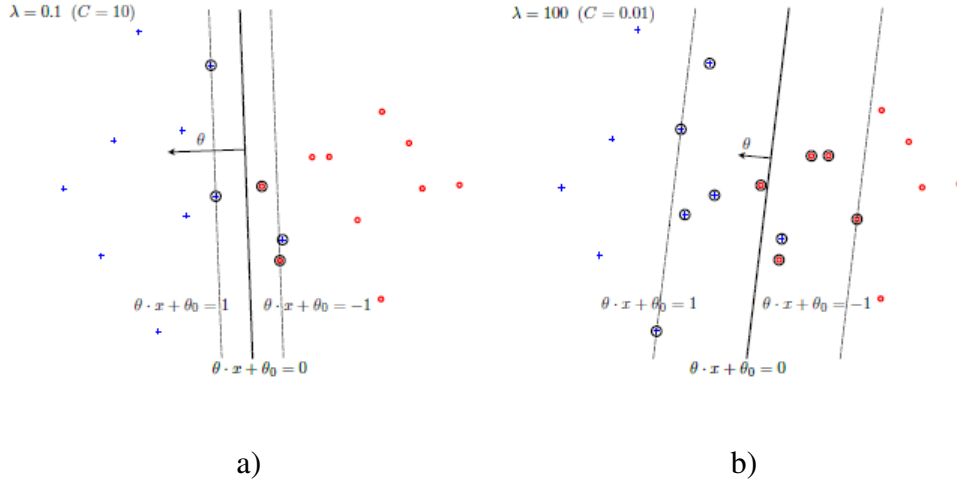


Figure 3: The effect of slack when examples are no longer linearly separable

Our current formulation of the support vector machines can be written in a familiar form. Indeed, the primal version is exactly the same as minimizing

$$\frac{\lambda}{2} \|\theta\|^2 + \sum_{t=1}^n \text{Loss}_h(y^{(t)}(\theta \cdot x^{(t)} + \theta_0)) \quad (8)$$

with respect to θ and θ_0 . Here $\text{Loss}_h(z) = \max\{1 - z, 0\}$ is the hinge loss. To see that these are

equivalent, let us define

$$\xi_t = \text{Loss}_h(y^{(t)}(\theta \cdot x^{(t)} + \theta_0)) = \max\{1 - y^{(t)}(\theta \cdot x^{(t)} + \theta_0), 0\} \quad (9)$$

Clearly, $\xi_t \geq 0$ and

$$\xi_t \geq 1 - y^{(t)}(\theta \cdot x^{(t)} + \theta_0) \quad \text{or} \quad y^{(t)}(\theta \cdot x^{(t)} + \theta_0) \geq 1 - \xi_t \quad (10)$$

In other words, the slack variables are simply encoding the hinge loss in the primal formulation.

How will the dual formulation change in light of the slack variables in the primal? We will simply limit how large the Lagrange multipliers α_t can become. In other words, the adversary “gives up” on trying to further satisfy some of the classification constraints. More formally,

$$\begin{aligned} \text{(dual)} \quad & \max \sum_{t=1}^n \alpha_t - \frac{1}{2} \sum_{t=1}^n \sum_{t'=1}^n \alpha_t \alpha_{t'} y^{(t)} y^{(t')} (x^{(t)} \cdot x^{(t')}) \\ & \text{subject to } 0 \leq \alpha_t \leq 1/\lambda, \sum_{t=1}^n \alpha_t y^{(t)} = 0 \end{aligned} \quad (11)$$

Note that the larger the value of λ is, i.e., the more we wish to increase the margin at the expense of the constraints, the smaller the resulting $\hat{\alpha}_t$ must be.

We have to be a bit more careful now in reconstructing the offset parameter θ_0 . Not all the support vectors, i.e., points with non-zero α_t 's, will lie exactly on the margin boundaries. Indeed, there will be points that lie on the wrong side of the margin boundaries or points that are misclassified altogether. The *complementary slackness* constraints that characterize the dual solution are now given by

$$\hat{\alpha}_t = 0 \Rightarrow y^{(t)} \left(\sum_{t'=1}^n \hat{\alpha}_{t'} y^{(t')} (x^{(t')} \cdot x^{(t)}) + \hat{\theta}_0 \right) \geq 1 \quad (\text{non-support vectors}) \quad (12)$$

$$\hat{\alpha}_t \in (0, 1/\lambda) \Rightarrow y^{(t)} \left(\sum_{t'=1}^n \hat{\alpha}_{t'} y^{(t')} (x^{(t')} \cdot x^{(t)}) + \hat{\theta}_0 \right) = 1 \quad (\text{SVs, on the margin}) \quad (13)$$

$$\hat{\alpha}_t = 1/\lambda \Rightarrow y^{(t)} \left(\sum_{t'=1}^n \hat{\alpha}_{t'} y^{(t')} (x^{(t')} \cdot x^{(t)}) + \hat{\theta}_0 \right) \leq 1 \quad (\text{SVs, margin violations}) \quad (14)$$

We can use the points for which $\hat{\alpha}_t$ lies in the interior of the possible values, i.e., $\hat{\alpha}_t \in (0, 1/\lambda)$, to reconstruct $\hat{\theta}_0$:

$$\hat{\theta}_0 = \frac{1}{|V'|} \sum_{t \in V'} \left\{ y^{(t)} - \left(\sum_{t'=1}^n \hat{\alpha}_{t'} y^{(t')} (x^{(t')} \cdot x^{(t)}) \right) \right\}, \quad V' = \{i | 1/\lambda > \hat{\alpha}_i > 0\}. \quad (15)$$

Learning Objective

You need to know:

1. What is the primal and dual form for SVM when we consider mis-classification errors
2. What is the definition of support vectors and how to check if a training instance is a support vector
3. What is the connection between SVM's training objective and hinge loss

A Deriving the Dual Problem for Soft-Margin SVM

The primal form of the problem in the notes is as follows:

$$\min \frac{\lambda}{2} \|\theta\|^2 + \sum_{t=1}^n \xi_t \quad (16)$$

$$\text{subject to } y^{(t)}(\theta \cdot x^{(t)} + \theta_0) \geq 1 - \xi_t, \xi_t \geq 0, \text{ for } t = 1, \dots, n \quad (17)$$

which is equivalent to the following:

$$\min \frac{1}{2} \|\theta\|^2 + C \sum_{t=1}^n \xi_t \quad (18)$$

$$\text{subject to } y^{(t)}(\theta \cdot x^{(t)} + \theta_0) \geq 1 - \xi_t, \xi_t \geq 0, \text{ for } t = 1, \dots, n \quad (19)$$

where $C = 1/\lambda$.

Okay, now let us focus on the last optimization problem. How do we optimize this problem? Remember we introduced Lagrangian multipliers to convert the original constrained optimization problem into a new problem. Here, similarly, we introduce the following Lagrangian:

$$\mathcal{L}(\theta, \theta_0, \xi, \alpha, \beta) = \frac{1}{2} \|\theta\|^2 + C \sum_{t=1}^n \xi_t + \sum_{t=1}^n \alpha_t (1 - \xi_t - y^{(t)}(\theta \cdot x^{(t)} + \theta_0)) + \sum_{t=1}^n \beta_t (-\xi_t) \quad (20)$$

where $\alpha_t \geq 0, \beta_t \geq 0$ for $t = 1, \dots, n$.

The optimization problem is now as follows:

$$\min_{\theta, \theta_0, \xi} \max_{\alpha \geq 0, \beta \geq 0} \mathcal{L}(\theta, \theta_0, \xi, \alpha, \beta) \quad (21)$$

Note that here θ, α, ξ and β are vectors. $\alpha \geq 0$ means all elements in the vector α are non-negative. This problem is in fact equivalent to:

$$\max_{\alpha \geq 0, \beta \geq 0} \min_{\theta, \theta_0, \xi} \mathcal{L}(\theta, \theta_0, \xi, \alpha, \beta) \quad (22)$$

This is the **dual problem** as we have discussed in class. Why do we do this? Because we hope that we can rewrite the following term as a function $f(\alpha, \beta)$, i.e., a function that involves only α and β . If we could do so, then we can just optimize such a function $f(\alpha, \beta)$ by tuning α and β :

$$\min_{\theta, \theta_0, \xi} \mathcal{L}(\theta, \theta_0, \xi, \alpha, \beta) \quad (23)$$

Okay. Now let's look at the above sub-problem. First of all, this is an optimization problem. α and β are now free variables, so we could regard them as "constants". Let's try to see what conditions we could have if the minimum is reached. Naturally if we take the following gradients they shall

be zeros when the minimum is reached:

$$\frac{\partial \mathcal{L}(\theta, \theta_0, \xi, \alpha, \beta)}{\partial \theta} = 0 \quad (24)$$

$$\frac{\partial \mathcal{L}(\theta, \theta_0, \xi, \alpha, \beta)}{\partial \theta_0} = 0 \quad (25)$$

$$\frac{\partial \mathcal{L}(\theta, \theta_0, \xi, \alpha, \beta)}{\partial \xi_t} = 0 \quad \text{for each } t \quad (26)$$

These conditions give us the following facts:

$$\theta - \sum_{t=1}^n \alpha_t y^{(t)} x^{(t)} = 0 \quad (27)$$

$$- \sum_{t=1}^n \alpha_t y^{(t)} = 0 \quad (28)$$

$$C - \alpha_t - \beta_t = 0 \quad \text{for each } t \quad (29)$$

Now, let's look at the $\mathcal{L}$ function again, and see what simplifications we can make based on the above facts:

$$\mathcal{L}(\theta, \theta_0, \xi, \alpha, \beta) \quad (30)$$

$$= \frac{1}{2} \|\theta\|^2 + C \sum_{t=1}^n \xi_t + \sum_{t=1}^n \alpha_t (1 - \xi_t - y^{(t)}(\theta \cdot x^{(t)} + \theta_0)) + \sum_{t=1}^n \beta_t (-\xi_t) \quad (31)$$

$$= \frac{1}{2} \|\theta\|^2 + \sum_{t=1}^n (C - \alpha_t - \beta_t) \xi_t + \sum_{t=1}^n \alpha_t (1 - y^{(t)}(\theta \cdot x^{(t)} + \theta_0)) \quad (32)$$

Since $C - \alpha_t - \beta_t = 0$ we can drop the 2nd term above:

$$\mathcal{L}(\theta, \theta_0, \xi, \alpha, \beta) \quad (33)$$

$$= \frac{1}{2} \|\theta\|^2 + \sum_{t=1}^n \alpha_t (1 - y^{(t)}(\theta \cdot x^{(t)} + \theta_0)) \quad (34)$$

$$= \frac{1}{2} \|\theta\|^2 + \sum_{t=1}^n \alpha_t - \sum_{t=1}^n \alpha_t y^{(t)}(\theta \cdot x^{(t)} + \theta_0) \quad (35)$$

$$= \frac{1}{2} \|\theta\|^2 + \sum_{t=1}^n \alpha_t - \sum_{t=1}^n \alpha_t y^{(t)}(\theta \cdot x^{(t)}) - \sum_{t=1}^n \alpha_t y^{(t)} \theta_0 \quad (36)$$

$$= \frac{1}{2} \|\theta\|^2 + \sum_{t=1}^n \alpha_t - \left(\sum_{t=1}^n \alpha_t y^{(t)} x^{(t)} \right) \cdot \theta - \left(\sum_{t=1}^n \alpha_t y^{(t)} \right) \theta_0 \quad (37)$$

Since $\theta = \sum_{t=1}^n \alpha_t y^{(t)} x^{(t)}$ and $\sum_{t=1}^n \alpha_t y^{(t)} = 0$ we have:

$$\mathcal{L}(\theta, \theta_0, \xi, \alpha, \beta) \quad (38)$$

$$= \frac{1}{2} \|\theta\|^2 + \sum_{t=1}^n \alpha_t - \theta \cdot \theta \quad (39)$$

$$= -\frac{1}{2} \|\theta\|^2 + \sum_{t=1}^n \alpha_t \quad (40)$$

$$= \sum_{t=1}^n \alpha_t - \frac{1}{2} \sum_{t'=1}^n \sum_{t=1}^n \alpha_t \alpha_{t'} y^{(t)} y^{(t')} x^{(t)} \cdot x^{(t')} \quad (41)$$

Okay, now we have reached a form that involves only α (well, we expected to have a form that involves both α and β , but it turns out there is no β in this form, which is even better). Now let's go back to our original problem. We would like to do the following:

$$\max_{\alpha \geq 0, \beta \geq 0} \min_{\theta, \theta_0, \xi} \mathcal{L}(\theta, \theta_0, \xi, \alpha, \beta) \quad (42)$$

$$= \max_{\alpha \geq 0, \beta \geq 0} \min_{\theta, \theta_0, \xi} \sum_{t=1}^n \alpha_t - \frac{1}{2} \sum_{t'=1}^n \sum_{t=1}^n \alpha_t \alpha_{t'} y^{(t)} y^{(t')} x^{(t)} \cdot x^{(t')} \quad (43)$$

Since the above form only involves α , we can remove all other variables, leading to the following problem:

$$\max_{\alpha} \sum_{t=1}^n \alpha_t - \frac{1}{2} \sum_{t'=1}^n \sum_{t=1}^n \alpha_t \alpha_{t'} y^{(t)} y^{(t')} x^{(t)} \cdot x^{(t')} \quad (44)$$

or :

$$\min_{\alpha} \frac{1}{2} \sum_{t'=1}^n \sum_{t=1}^n \alpha_t \alpha_{t'} y^{(t)} y^{(t')} x^{(t)} \cdot x^{(t')} - \sum_{t=1}^n \alpha_t \quad (45)$$

What are the constraints? Here they are:

$$\alpha_t \geq 0, \beta_t \geq 0 \quad (46)$$

for $t = 1, \dots, n$, and (from Equation 17, which involves α only):

$$\sum_{t=1}^n \alpha_t y^{(t)} = 0 \quad (47)$$

However the objective does not involve β , and $\beta_t = C - \alpha_t$. This means the second set of constraints are really $C - \alpha_t \geq 0$, or $\alpha_t \leq C$ for all t . This concludes that the constraints are:

$$0 \leq \alpha_t \leq C \text{ for all } t = 1, \dots, n \quad (48)$$

$$\sum_{t=1}^n \alpha_t y^{(t)} = 0 \quad (49)$$

Okay. Finally here comes our dual form for the SVM with soft-margin:

$$\min_{\alpha} \frac{1}{2} \sum_{t'=1}^n \sum_{t=1}^n \alpha_t \alpha_{t'} y^{(t)} y^{(t')} x^{(t)} \cdot x^{(t')} - \sum_{t=1}^n \alpha_t \quad (50)$$

subject to :

$$0 \leq \alpha_t \leq C \text{ for all } t = 1, \dots, n \quad (51)$$

$$\sum_{t=1}^n \alpha_t y^{(t)} = 0 \quad (52)$$

This is in fact a standard quadratic programming problem. There are standard algorithms for solving such a quadratic programming problem. In fact, in the case of linear SVM, there exists very efficient algorithms such as Sequential Minimal Optimization (SMO).